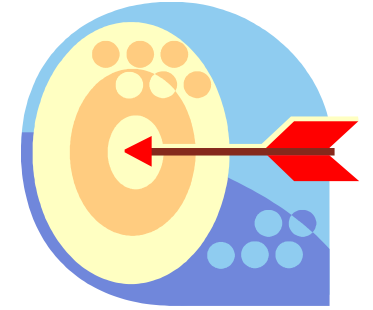


Lesson 7: Introduction to Service Management

Lesson Objectives



This lesson has been designed to enable you to:

- List and explain the basic service types
- Explain how to create a service configuration
- Explain Bootstrapping
- Describe how the File System adapter works
- Describe how On-Fault Processing works
- Explain how the Command Line adapter functions
- Describe and demonstrate the use of a Lightweight JDBC adapter with a code list
- Describe how the Invoke Business Process service works

Introduction

What makes Sterling B2B Integrator so flexible is the capabilities of its services. A service is a small program that B2Bi can call to accomplish tasks in a bp. Usually, the high-level purpose of a service is to achieve some pre-defined type of integration activity, such as:

- Communicating with external applications or middleware (by way of special services called adapters)
- Performing data manipulations, such as translation, transformation, splitting, and joining
- Routing data based on payload, content, or meta data
- Publishing data to interested subscribers, which can trigger a new business process instance or allow an executing process to continue
- Executing one or more B2B protocols
- Spawning a pre-defined business process (nested process)

Services that communicate with external systems are called **adapters**. Input Services and Output Services are adapters. A service that runs inside Sterling B2B Integrator and does not touch outside systems will retain the **service** title in its name.

Services

Consider that a business process involves determining how many open orders a specific customer might have. The activities might include:

- Looking up the customer number
- Retrieving all orders for that number from an orders database
- Removing all but pending orders from that table
- Reporting the information in printed form

Each of these activities could be accomplished through a service.

Adapters

Adapters can appear at any point of a business process and are often used to initiate business processes. For example, a File System Adapter could poll a directory to detect any new files (purchase orders) and initiate a business process that processes the input.

Service Types

These are the basic service types in Sterling B2B Integrator:

- **Internal Services** - that are totally inside Sterling B2B Integrator. While internal services accept parameters and produce results, they do not directly interact with outside systems. These types of services will usually retain “Service” in their name. For example the Translation Service.
- **Input Services** - that receive data from outside systems.
- **Output Services** - that send data to outside systems.

Input and output services are collectively known as adapters. Adapters interact with the world outside of the Sterling B2B Integrator J2EE environment.

Service Configurations

- The services you include in your business process model may require configuration in order for your process to run correctly. You configure some parameters in Sterling B2B Integrator interface (this is called creating a service configuration), some in either the interface or the GPM, and some exclusively in the GPM. A few require manual editing in the BPML code.
- The reference documentation for each service describes the parameters for the service and indicates where to perform the parameter configuration.
- Generally, you can configure service parameters either during initial service configuration in Sterling B2B Integrator interface, or during business process creation using the GPM. You can modify service configuration parameters at any time.

File System Adapter

Overview

The File System adapter (FSA) is functionally the simplest of all the Sterling B2B Integrator adapters. It collects files from a disk and can be used to initiate business processes. It also extracts files from Sterling B2B Integrator back to a disk. It can retrieve and send any type of file, but it does not actually read the file.



How the File System Adapter Works

- Use the File System adapter to collect (import) files from a file system into a business process and extract (export) files from a business process to a file system. You can configure the File System adapter to start a business process after files are collected from the file system or include the File System adapter in a business process flow. In addition, you can schedule the File System adapter to run at specific time intervals.
- You can set a file system adapter configuration to collect, extract, or both. However, each instance of a File System adapter in a business process will either **collect or extract**, it can not do both at the same time. Normally, you should use only the extraction side of the File System adapter in a business process as the collection is usually done by a bootstrapped configuration designed to kickoff a bp rather than being part of one.
- The FSA works with the file system of the server where the configuration is active. In a clustered environment the adapter can only run on a specific node, unless there is a type of shared file storage. Other file systems can be used but they must be mounted as part of the local system

Collection Vs. Extraction

Collection

When a FSA is set to Collection there are two types of modes, each one causing different behavior:

- Bootstrapped - Is designed to have the FSA monitor a folder based on a schedule and start a business process when a file is collected.
- Non-Bootstrapped - Is designed to be part of a business process and collect a file as part of the bp flow.

Extraction

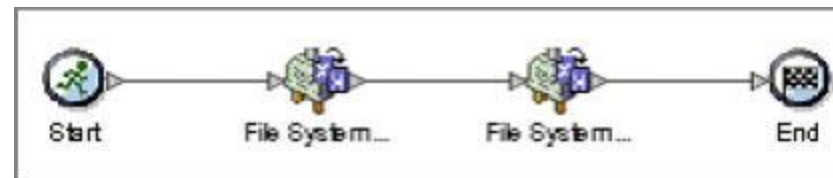
- Extracts the data to the specified folder. The file that is created is given with the specified name.
- When a FSA is used to extract a file to disk in a bp it will extract Primary Document. The adapter can be configured to use the original name of the file picked up prior or have the adapter create a new name based on a setting or variables.

File System Adapter Configuration



After you install the adapter, you need to create an instance configuration in Sterling B2B Integrator before you can use the adapter.

Please pause this presentation review the steps for configuration in your student manual.



File System Adapter Configuration (continued)

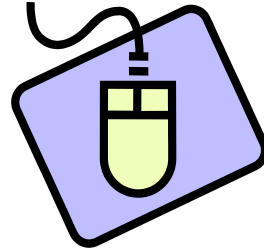
When the file system adapter is placed into a business process, you must select a configuration instance. The following table describes the fields used to configure the File System adapter in the GPM. This table contains only the fields that are configured in the GPM. The values in parentheses represent the corresponding BPML values.

Service Editor-File System Adapter		
Name	File System Adapter	
Config.	AS3F5Adapter	
Message To Service Message From Service		
Output Msg	Obtain Process Data first, then Messages	
Message Name	FileSystemInputMessage	
Name	Value	Use XPATH?
fileModTimeThreshold		☐
checkDelete		☐
appendOnExtract		☐
noFilesSetSuccess		☐

File System Adapter Configuration (continued)



Please Pause this presentation and review the fields and descriptions that are listed in your book for the configurable parameters in your student manual



Exercise 7–1: File System Adapter

Your turn!

Pause this presentation, read the Scenario, then follow the instructions for Exercise 7-1

Bootstrapping

Bootstrapping

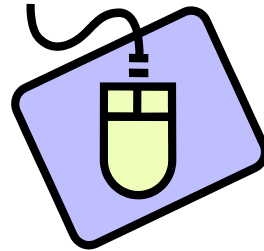
As discussed before, business processes may be scheduled to run at a given date and time. However, the trigger for a business process may be the arrival of data from outside Sterling B2B Integrator, such as a JMS queue. **Bootstrapping is defined as the act of initiating a business process.** If the Sterling B2B Integrator Scheduler bootstraps a business process, no outside activity is needed, although the business process may, and probably will, make use of external systems in the course of its operation. Some adapters may be configured to bootstrap a business process when they receive data.

Here are some additional facts about bootstrapping:

1. There may be a many-to-many relationship between bootstrapping adapters and business processes.
2. Many adapters may be configured to start the same business process.
3. The business process bootstrapped may be selected dynamically.
4. The bootstrapping adapter may be at the beginning of a business process or may occur in the middle of a business process.
5. When creating a service to bootstrap, a system generated business process is built that uses the service by way of the Scheduler. The business process is normally called `Schedule_<service_configuration_name>`.

Bootstrapping

- Sterling B2B Integrator includes several bootstrap adapters – adapters that can start processes dynamically without being part of the process model. When building your business process models, do not include these adapters as the first step in a business process model. Using bootstrap adapters in this way is preferable because your goal is to reuse components whenever possible, for easier maintenance of your process models. If you create an adapter configuration specific to the business process, you are less likely to be able to reuse the adapter configuration to start other business processes.
- Bootstrapping allows you to setup the ability for triggers to kickoff business processes. Such as a Filesystem adapter scheduled to check a folder every minute. When a file is found it will bootstrap or start a new business process to process the collected file.
- Bootstrapped adapters can be set to only kickoff a specific business process or dynamically lookup a process allowing a single adapter instance to start multiple business process based off of a key value such as filename.



Exercise 7–2: Modify the File System Adapter

Exercise 7–3: Edit the File System Adapter

Your turn!

Pause this presentation, read the Scenario, then follow the instructions on Exercise 7-2 and 7-3

OnFault Processing

OnFault Processing

The OnFault element is used to handle errors. You can include OnFault elements in any complex activity for which it may be necessary to recover from faults so that the process can continue. The OnFault element contains a fault handling activity.

OnFaults are commonly tied to sequences. This could be a sequence that covers the entire business process flow or a smaller sequence such as a choice or repeating loop. Each OnFault can also be setup to trigger on different error conditions. This could be specific error or setup to be generic and trigger on any error.

For each OnFault condition you build a subroutine to run if an error condition is met. It is common to build some type of alert bp like email people or a file drop off bp and then invoke those bp's through the subroutine. This way you can reuse common bp's instead of building the steps into every OnFault.

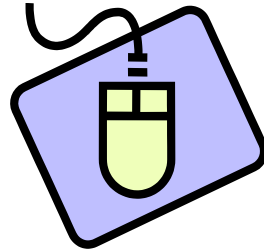


SMTP Send Adapter

The SMTP Send adapter allows documents to be sent to any valid e-mail address using an accessible (SMTP-aware) mail server, usually the mail server of the trading partner that is running Sterling B2B Integrator. It does this by enabling Sterling B2B Integrator to mail (send) documents using SMTP, within Sterling B2B Integrator, to the designated mail server. The back-end mail server system examines the document and does further processing, including the actual sending of the e-mail. This adapter can be used when messages and documents need to be sent during the execution of a business process.

The SMTP Send adapter can mail (send) documents to the back-end mail server of trading partners. The document is then e-mailed to the recipient designated in the To field of the configuration. This adapter picks up the document sent from the primary document in the business process.





Exercise 7–4: OnFault Processing Using SMTP Send Adapter

Your turn!

Pause this presentation, read the Scenario, then follow the instructions on Exercise 7-4

Command Line 2 Adapter

Command Line 2 Adapter

The Command Line 2 adapter is a second generation adapter that enables Sterling B2B Integrator to run a program from a command line in a business process. This includes executable programs, scripts, or operating system (OS) commands external to Sterling B2B Integrator. The Command Line 2 adapter only operates in a remote implementation. This does not necessarily mean that it has to run remotely. The Command Line 2 adapter runs in a separate JVM (Java Virtual Machine), which may be on the machine where Sterling B2B Integrator is installed or on a remote machine. The Command Line 2 adapter supports large files up to 12 GB and provides better memory allocation than the Command Line adapter. The Command Line 2 adapter will eventually replace the Command Line adapter.



Caution

Use the Command Line 2 adapter only for non-interactive programs that run from the command line. This adapter is not intended to work with browser-based or other Graphical UI-based programs. Such programs introduce an interactive element into a business process, which is typically set up to be automated.

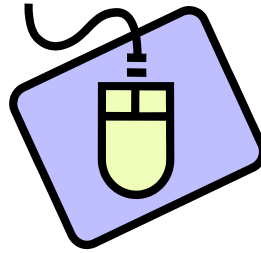
Command Line 2 Adapter (continued)

The following steps summarize how the Command Line 2 adapter is typically used in a business process:

1. The adapter writes the content of the current primary document to a file in the working directory specified as the value of the working directory parameter. The name of this file is specified by the value of the inputFile parameter.
2. Sterling B2B Integrator runs an executable program that picks up the file and sends it to the legacy system.
3. The legacy system returns a file, which now includes the customer billing information, and the adapter retrieves it. The file returned is specified by the value of the outputName parameter.
4. The adapter reads the file contents into the primary document.
5. Sterling B2B Integrator performs the next operation in the business process.



Implementing the Command Line 2 Adapter



You can implement a Command Line 2 adapter to do the following:

- Execute commands using the command line from within a business process.
- Invoke the Command Line 2 adapter on a schedule and then start a new business process using the output from the adapter.

Important!!! This can be used if you want to schedule a command line program that accesses a legacy database on a regular schedule and then uses the output in a business process.

Your Turn!

Please Pause this presentation and follow the instructions that prepare you for Exercise 7-5 and then proceed thru Exercise 7-5 in your student manual.

Invoke Business Process Service

The Invoke Business Process Service

The Invoke Business Process Service is used to start an instance of a subprocess (child) from within the current instance of the business process (parent).

Note: The Invoke Business Process Service is also referred to as the **Invoke Sub-process Service**.

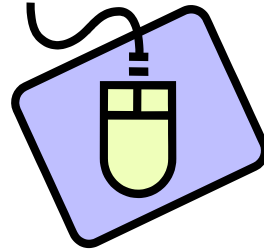
The Invoke Business Process Service, when used, instantiates an instance of a subprocess, passing the current workflow context and the primary document to the new instance.

- When the Invoke Business Process Service is set to synchronous mode, the parent suspends processing until it receives data from the child. In synchronous mode, the parent is notified when the child encounters errors.
- When the Invoke Business Process Service is set to asynchronous mode, the parent and child process data simultaneously and independently of each other. Therefore, the parent does not receive notification when the child encounters errors.
- When the Invoke Business Process Service is set to run a subprocess inline, the subprocess runs as part of the parent process, sharing the same process data.
- When the Invoke Business Process Service is set to run in embedded mode, the subprocess runs with no persistence, meaning that no record of the process is recorded in Sterling B2B Integrator and no tracking is done.

Implement Invoke Business Process Service

The Invoke Business Process service does not require you to build a configuration as all of the configuring is done in the GPM/BPML. If you do build a configuration all that is entered is a name and description. But it can then be used as any other configuration. Commonly developers just use the pre-built **InvokeBusinessProcessService** configuration and configure in the GPM/BPML.





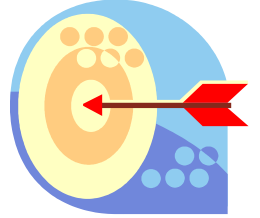
Exercise 7–6: Invoke Business Process Service

Your turn!

Pause this presentation, read the Scenario, then follow the instructions on Exercise 7-6

Lightweight JDBC Adapter

Lightweight JDBC Adapter



The Lightweight Java Database Connectivity (JDBC) adapter enables you to retrieve data from a JDBC-compliant database or update a JDBC-compliant database as part of a business process within Sterling B2B Integrator, using one of the following methods:

- XPath queries for dynamic SQL queries (where the definition of the SQL statement is not known when the business process is defined)
- Constants, when lookups are not necessary

Lightweight JDBC Adapter (continued)

Unlike the JDBC adapter, the Lightweight JDBC adapter does not use a map, where the SQL statements are fixed. The Lightweight JDBC adapter enables you to specify parameters in the SQL statement that are supplied at run time.

The Lightweight JDBC adapter executes a query, and you receive an XML document that contains the results.

Then you can either:

- Load the XML document into process data.
- Carry the XML document forward into another service in the business process.

Use the Lightweight JDBC adapter if you simply want to get data, and need the flexibility of specifying various SQL queries. Use the JDBC adapter if you plan to always execute the same SQL statement, or if you have complex output and you want to manipulate the data from a database and control the structure of the output.



How the Lightweight JDBC Adapter Works

The Lightweight JDBC adapter can start a business process, or it can be used in the middle or at the end of a business process. This service can be scheduled to run at weekly or timed intervals.

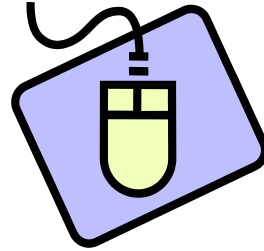
The following steps summarize how the Lightweight JDBC adapter might be used in a business process:

1. Using the parameters specified in the Graphical Process Modeler for the Lightweight JDBC adapter configuration, the adapter executes a query.
2. The adapter receives the query results from an SQL database query.
3. The adapter generates a primary document that contains the query results in XML format.
4. The adapter sends the query results to the next step in the business process.

Important!!!! If you want to use XPath queries, you will first need to load the data these queries will reference into the process data.

Codelists

- JDBC connections between Sterling B2B Integrator and outside databases are very powerful, useful tools. However, creating that connection can be complicated. For some tasks, you may not need such advanced functionality. In these cases, you can make use of the Sterling B2B Integrator function; the code list.
- Code lists are created as tables through the Sterling B2B Integrator UI, but are held as records within the Sterling B2B Integrator database, in a table named `codelist_xref_item`. These records can be accessed through JDBC adapters in your business process, and also through maps you create. Like any other table, you can select, update, and insert records, and call stored procedures or functions when working with code lists.



Exercise 7–7: Creating a CodeList

Your turn!

Pause this presentation, read the Scenario, then follow the instructions on Exercise 7-8

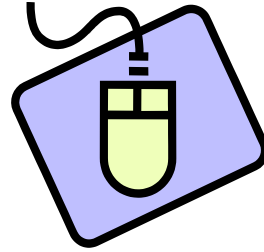


Sterling B2B Integrator Lightweight JDBC Adapter Configuration

Sterling B2B Integrator Configuration



Please pause this presentation and review the table that describes the fields used to configure the Lightweight JDBC adapter in Sterling B2B Integrator in your student manual



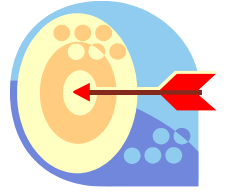
Exercise 7–8: Lightweight JDBC Adapter

Your turn!

Pause this presentation, read the Scenario, then follow the instructions on Exercise 7-8

Wait Service

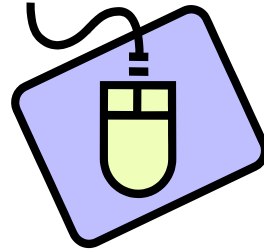
Wait Service



The Wait service allows a business process to wait for a period of time before moving on to the next step in the process. The Wait service does not hold its active thread during this wait time, which can reduce resource consumption.

The Wait service performs a function similar to the Sleep service: it allows you to pause a business process for a set length of time. However, unlike the Sleep service, the Wait service does not retain an active processing thread, which frees the thread for other processing jobs. Use the Wait service when you need to put a business process in a Wait state for at least one minute. If you need to put a business process in a wait state for less than a minute, use the Sleep service.

Because a configuration of the Wait service is installed with Sterling B2B Integrator, no configuration is necessary.



Exercise 7–9: Wait Service

Your turn!

*Pause this presentation, read the Scenario, then
follow the instructions on Exercise 7-9*



Congratulations !!!!!

Completed Lesson Objectives -

This lesson was designed to enable you to:

- List and explain the four basic service types
- Explain how to create a service configuration
- Explain Bootstrapping
- Describe how the File System adapter works
- Describe how OnFault Processing works
- Explain how the Command Line adapter functions
- Describe and demonstrate the use of a Lightweight JDBC adapter
- Describe how the Invoke Business Process service works